

1. Administrative & Study Identification

Full Study Title: A Study of Multiple Immunotherapy-Based Treatment Combinations in Patients With Locally Advanced Unresectable or Metastatic Gastric or Gastroesophageal Junction Cancer (G/GEJ) or Esophageal Cancer (Morpheus-Gastric and Esophageal Cancer) **Short Title/Acronym:** MORPHEUS-Gastric and Esophageal Cancer (MORPHEUS-GC / MORPHEUS-EC) **Protocol Number:** Y039609 (EudraCT: 2016-004529-17) **NCT Number:** NCT03281369 **Posting ID:** 502915919806106364 **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Sponsor/Company:** Roche (Hoffmann-La Roche) **Investigational Product:** Atezolizumab (Tecentriq) in combination with novel targeted agents (e.g., Tiragolumab, Linagliptin, PEGPH20, Cobimetinib, BL-8040) and/or standard chemotherapy. **Phase of Development:** Phase 1 / Phase 2 (Phase Ib/II) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of multiple immunotherapy-based treatment combinations compared with active control (standard-of-care chemotherapy) in patients with locally advanced unresectable or metastatic gastric, gastroesophageal junction (G/GEJ), or esophageal cancer. **Secondary Objectives:** To evaluate investigator-assessed Progression-Free Survival (PFS), Overall Survival (OS), and landmark survival rates of the experimental treatment combinations.

2.2 Background & Rationale To address the poor clinical outcomes in patients with previously untreated or treated locally advanced unresectable/metastatic G/GEJ or esophageal cancer. While chemotherapy and immune checkpoint inhibitors (ICIs) are standard-of-care, response rates remain variable. The MORPHEUS open-label umbrella platform aims to identify early efficacy and safety signals of novel immune-based combinations (such as dual blockade of TIGIT and PD-L1 using tiragolumab + atezolizumab) to enhance anti-tumor immune responses, overcome immune escape mechanisms, and improve survival.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Study Platform) **Control Type:** Active Comparator (Standard-of-care chemotherapy) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment: 410 participants across multiple cohorts. **Number of Sites:** Multinational across multiple countries (including USA, UK, Australia, Israel, South Korea, Taiwan). **Estimated Study Start:** 13-Oct-2017 **Estimated Primary Completion:** 09-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed diagnosis of squamous cell carcinoma or adenocarcinoma of the esophagus, gastric, or GEJ in locally advanced unresectable or metastatic disease. 2. First-line (1L) cohorts: No prior systemic treatment for their advanced disease. Second-line (2L) cohorts: Progression during or following a 1L platinum- or fluoropyrimidine-containing chemotherapy regimen. 3. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0, 1, or 2.

4.2 Exclusion Criteria 1. Pregnancy or breastfeeding, or intention of becoming pregnant during study treatment or within 5 months after the final dose of atezolizumab/investigational products. 2. Active, untreated, or symptomatic central nervous system (CNS) metastases. 3. Lack of agreement to remain abstinent or use highly effective contraceptive measures (including refraining from donating eggs/sperm).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Atezolizumab 1200 mg intravenously (IV) every 3 weeks (21-day cycles) in combination with assigned experimental therapies (e.g., Tiragolumab 600 mg, Linagliptin, PEGPH20) and/or standard chemotherapy (e.g., Cisplatin [Platinol, ATC: L01XA01] + 5-FU, Ramucirumab + Paclitaxel, mFOLFOX6). **Route of Administration:** Intravenous (IV) for checkpoint inhibitors and chemotherapy; Oral (PO) for certain targeted agents (e.g., linagliptin). **Duration of Treatment:** Treatment in 21-day cycles until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease-related symptoms and toxicities. **Prohibited:** Concurrent administration of other investigational anti-cancer therapies outside the protocol framework.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline Imaging (CT/MRI), Labs Baseline

Day 1 | Randomization, First Dose of Study Treatments | | **Interim**
| Every 3 Weeks (Day 1 of Cycles) | Dosing, Safety Labs, AE
Assessment; Tumor Imaging every 6-9 weeks | | **End of Study** | 30
Days Post-Treatment | Final Safety Assessment, Survival Follow-up
|

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Objective Response Rate (ORR):**

Percentage of participants with Objective Response, as determined by investigator according to Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1. 2. **Safety:** Percentage of participants with Adverse Events (AEs). 3. **Arm-Specific Safety:** For Arm 1L-A, Percentage of participants with Serious and Non-serious Treatment-related AEs.

7.2 Secondary & Exploratory Endpoints 1. **Progression-Free**

Survival (PFS): As determined by investigator according to RECIST v1.1. 2. **Overall Survival (OS).** 3. **Landmark Survival:** Percentage of participants who are alive at Month 6 and at Month 12.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection and grading of AEs using NCI CTCAE version 4.0/5.0. **Serious Adverse Events (SAEs):** Reported continuously, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal platform review committee analyzing safety and risk-benefit across multiple MORPHEUS arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy outcomes; Safety-evaluable population for toxicity. **Sample Size Justification:** Powered for early signal detection (Phase 1b/2 proof of concept). Small comparative cohorts evaluated sequentially rather than definitive Phase 3 superiority sizing. **Handling of Missing Data:** Standard survival analysis censoring logic for PFS/OS endpoints (e.g., censoring at the last date the patient was known to be alive and progression-free).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits to ensure data integrity and protocol adherence. **Audit Trail:**

Detailed electronic case report form (eCRF) locking and data clarification procedures.

11. Study Identifiers & Governance

Primary ID: Y039609 **Registry Number:** NCT03281369 **Posting ID:** 502915919806106364 **Sponsor/Lead Organization:** Roche **Study Title:** MORPHEUS-Gastric and Esophageal Cancer **Official Regulatory Title:** A Phase Ib/II, Open-Label, Multicenter, Randomized, Umbrella Study Evaluating the Efficacy and Safety of Multiple Immunotherapy-Based Treatment Combinations in Patients With Locally Advanced Unresectable or Metastatic Gastric or Gastroesophageal Junction Cancer or Esophageal Cancer (Morpheus-Gastric and Esophageal Cancer)

12. Clinical Framework (Oncology Specific)

Preferred UMLS Name: Upper Gastrointestinal Cancer **Primary Condition / Disease Areas:** Gastric Adenocarcinoma or Gastroesophageal Junction Adenocarcinoma or Esophageal Carcinoma **Diagnosis Threshold:** Histologically/cytologically confirmed locally advanced unresectable or metastatic disease. **Histology Designations:** Adenocarcinoma (MeSH: D000230 | SNOMED ID: 1187332001) **Medical Methodology:** Multiple Immunotherapy-Based Treatment Combinations (Umbrella platform). **Optimization Target:** Maximizing anti-tumor immunity, ORR, and extending PFS/OS.

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway incorporating checkpoint inhibitors and targeted combinations. **Education Protocol (HCP):** Investigator site training on specific administration protocols for various umbrella study arms. **Education Protocol (Patient):** Informed consent education on experimental combinations and potential immune-related AEs. **Quality Audit Mechanism:** Radiological RECIST v1.1 review tracking and investigator compliance auditing.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until death or study closure (expected to span multiple years). **Onsite Visit Cadence:** Every 3 weeks (21-day cycles). **Remote Monitoring Frequency:** Continuous safety monitoring. **Checklist Review Interval:** Tumor assessments interval every 6 to 9 weeks depending on treatment phase.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					